Anand Kumar

Quantum Communications · FSO & Atmospheric Channel Modeling · Satellite/NTN Systems

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Education

Guru Gobind Singh Indraprastha University, New Delhi, India *2024 – 2028*
Bachelor of Technology, Electronics and Communication Engineering CGPA: 8.7/10.0

- Relevant coursework: Signals and Systems, Communication Systems, Analog Electronics, Digital Electronics, Network Theory, Engineering Mathematics
- Advanced training: Admitted Live Auditor, CMU 11-785 Deep Learning (graduate-level, synchronous Zoom); MIT 8.04 Quantum Physics I (OpenCourseWare)

Research Experience

IITM C-DOT Samgnya Technologies Foundation (National Quantum Mission), IIT Delhi *Jun. 2026 – Present*
Research Intern, under Dr. Neel Kanth Kundu, Centre for Applied Research in Electronics, IIT Delhi

- Investigating Twin-Field QKD (TF-QKD) over turbulent Free-Space Optical (FSO) channels; adopted the Curty–Azuma–Lo (CAL) Protocol-3 asymptotic, collective-attack security model as the baseline and extended it to atmospheric links by replacing the fixed-loss channel with a log-normal turbulence-fluctuating transmittance
- Derived a finite-order analytical Hermite-moment expansion in the Gaussian log-transmittance domain, combined with an exact Euler–Stirling transformation back to the physical transmittance domain, to compute the turbulence-averaged X-basis error rate, Z-basis phase-error rate, and secret-key rate without repeated Monte Carlo or quadrature evaluation at each operating point
- Cross-validated two independently derived formulations of the expansion to better than 10^{-9} relative agreement across 78 checks, and against direct numerical Gaussian quadrature over a distance/turbulence sweep, keeping the fourth-order secret-key-rate approximation within 0.4% of the numerical reference
- Derived a zero-fitted-parameter matched-asymptotic surrogate for the CAL error-rate and key-rate quantities, and validated the resulting closed-form turbulence average against 2×10^5-sample Monte Carlo simulation across eleven operating points, achieving relative error under 3%
- Related project: TF-QKD / FSO Quantum Communication Research Framework (see Projects)

Scientific Analysis Group (SAG), Defence Research and Development Organisation, New Delhi *Jun. 2026 – Aug. 2026*
Research Intern, under Mr. Rajesh Kumar, Scientist 'E'

- Conducted three independent SNS-TF-QKD studies – a physical-layer OptiSystem prototype, a Python classical-processing pipeline, and QSAFire, an independent satellite-link simulation framework – reported and validated separately by design
- Built a three-station SNS-TF-QKD physical-layer prototype in OptiSystem (Alice/Bob/Charlie, 40 km SMF-28 fibre spans, 10 Gbit/s), implementing window-selection, send/not-send, decoy-intensity, and phase-randomisation protocol logic as MATLAB co-simulation blocks with bit-index-synchronised CSV logging across all three stations
- Built QSAFire, a Python satellite-link performance framework combining J2-perturbation orbital geometry, elevation-dependent slant range, atmospheric loss, Gamma-Gamma turbulence, pointing loss, and a fully asymmetric, jointly-optimised SNS-TF-QKD key-rate model applied to six Indian ground sites and four candidate links; validated the orbital solver and elevation–slant-range formulas against Landsat-8/9, Sentinel-2, and published Micius in-orbit data to within 1–3%
- Identified a plane-wave-vs-spherical-wave Rytov-variance error in QSAFire's uplink turbulence model, corrected it with a literature-anchored conversion ratio, and confirmed the fix by executing the corrected framework end to end
- Related project: QSAFire: Satellite-to-Ground SNS-TF-QKD Simulation Framework (see Projects)

Bharati Vidyapeeth's College of Engineering, New Delhi *Jul. 2025 – Present*
Undergraduate Research Assistant, under Dr. Yugnanda Malhotra

- Leading undergraduate research in Quantum Key Distribution (QKD), Free-Space Optical communication, and satellite quantum networking
- Developed climatology-aware atmospheric channel models (ERA5/CAMS reanalysis data) for turbulence, attenuation, and link availability across Indian quantum-communication corridors
- Designed simulation frameworks for BB84, decoy-state BB84, and entanglement-based QKD protocols over terrestrial and satellite FSO channels
- Published first-author research in Springer and contributed to book chapters and conference proceedings on satellite quantum-communication architectures

Projects

Decoy-State BB84 Satellite QKD Constellation Simulator *2025*

- Designed and simulated a satellite-based decoy-state BB84 QKD constellation, modeling orbital visibility, free-space channel attenuation, detector characteristics, and secure key generation across varying altitudes, link distances, and atmospheric losses; published as first author in Springer (see Publications)

TF-QKD / FSO Quantum Communication Research Framework *2026 – Present*
GitHub

- Modular open-source Python framework implementing the Curty–Azuma–Lo (CAL) Protocol-3 TF-QKD security proof: X/Z-basis gain and QBER calculations, cat-state phase-error estimation, asymptotic secret-key-rate pipeline, and PLOB repeaterless-bound comparison
- Extended to a finite-decoy linear-programming framework (CVXPY-based) for finite-decoy yield and phase-error bounds, with convergence, feasibility, and finite-vs-infinite-decoy consistency diagnostics
- Built a deterministic and log-normal-turbulence Free-Space Optical (FSO) channel model (Beer–Lambert atmospheric attenuation, Gaussian beam propagation, Rytov variance, Monte Carlo channel realizations) for ergodic secret-key-rate estimation under atmospheric turbulence
- Developing an analytical log-normal averaging framework (quadrature-based ergodic SKR) to replace expensive Monte-Carlo-only averaging while retaining it for validation; documented with MkDocs
- Related experience: Research Intern, IIT Delhi – TF-QKD over turbulent FSO channels (see Research Experience)

Climate-Aware Quantum Channel Modeling for Satellite QKD *2026*
GitHub

- Developed a physics-consistent satellite-to-ground optical channel model using ERA5 meteorological and CAMS aerosol reanalysis data, modeling Rayleigh scattering, aerosol extinction, atmospheric absorption, turbulence, and geometric beam-spreading losses
- Integrated these atmospheric effects into decoy-state BB84 simulations to evaluate QBER and secure key rate, benchmarking wavelength-dependent performance across geographically diverse Indian ground stations
- Related experience: Undergraduate Research Assistant, BVCOE – climatology-aware atmospheric channel modeling (see Research Experience)

QSAFire: Satellite-to-Ground SNS-TF-QKD Simulation Framework *2026*
GitHub

- Independent Python framework combining J2-perturbation orbital geometry, atmospheric transmittance, slant-path Rytov-variance turbulence, pointing/acquisition/tracking losses, and a fully asymmetric, jointly-optimised SNS-TF-QKD key-rate calculation across six Indian ground sites and four candidate links (developed at SAG, DRDO)
- Related experience: Research Intern, DRDO SAG – SNS-TF-QKD studies (see Research Experience)

SateSim: Satellite Communication Link-Budget Simulator *2026*
GitHub

- Built a satellite-to-ground communication simulator integrating SGP4/Skyfield orbital propagation with RF

link-budget analysis (FSPL, Friis transmission, Doppler shift, elevation-based visibility); computed SNR/BER for BPSK/QPSK modulation and delivered results via an interactive Streamlit dashboard with automated CSV/PDF reporting

CMU 11-785 – Paddy Microclimate Stress Classifier *2025*
GitHub

– Designed a BiLSTM–LSTM model for 3-class paddy stress forecasting across 6 Indian agroclimatic zones using 48-hour weather sequences (temperature, RH, VPD, ET0, soil moisture)
– Achieved 73.4% accuracy, Macro-F1 0.736, and 0.924 ROC-AUC under strict Leave-One-Location-Out generalisation to unseen coastal tropical climates; open-sourced on GitHub with post-hoc attention analysis

Research Publications & Presentations

– Kumar, A., Malhotra, Y., Gupta, A., Miglani, V., and Parikh, J., "Surrogate-Assisted Bayesian Optimization for Real-Time Parameter Adaptation in NPP-TF-QKD," accepted for presentation at the 2026 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS), IIT Roorkee, India, Dec. 2026.
– Kumar, A., Malhotra, Y., Gupta, A., and Parikh, J., "Design and Simulation of a Decoy-State BB84 Satellite QKD Constellation for Indian Ground Stations," *International Journal of Information Technology*, Springer Nature, 2025. DOI: 10.1007/s41870-025-02860-y
– Kumar, A. and Gupta, A., "Quantum Key Distribution over Free-Space Optical Channels," IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS), Dec. 2025.
– Kumar, A., Gupta, A., Malhotra, Y., and Parikh, J., "Quantum Key Distribution Beyond 5G and 6G: Hybrid Integrations, Testbeds, and Future Directions," in *Quantum Machine Learning: Concepts, Algorithms, and Applications*, CRC Press / Routledge, 2026.
– Kumar, A., Malhotra, Y., Puri, H., Parikh, J., and Talwar, R., "Climatology-Aware Satellite Optical Channel Model for Slant-Path Propagation," poster presentation, Optica Frontiers in Optics + Laser Science (FiO+LS) 2026, Joint Poster Session I, Rochester, NY, Sept. 2026.

Honors & Awards

– Selected for the Optica Student Leadership Conference 2026 in Rochester, New York, USA, held alongside Frontiers in Optics + Laser Science (FiO+LS); awarded competitive travel, accommodation, and conference funding by Optica

Technical Skills

– *Programming:* Python, MATLAB, C, C++
– *Quantum & Optical:* QKD, TF-QKD, SNS-TF-QKD, BB84, Decoy-State BB84, BBM92, FSO Communication, Satellite QKD
– *Scientific Computing:* NumPy, SciPy, Pandas, Matplotlib, PyTorch, Scikit-learn, Numerical Quadrature, Monte Carlo Simulation
– *Satellite & Simulation Tools:* CST Studio Suite, OptiSystem, SGP4, Skyfield, Streamlit, LaTeX, Git/GitHub

Leadership & Activities

– *President, BVP Optica (Optics & Photonics Society), BVCOE*, 2026 – Present – organised workshops and research sessions in optics, photonics, and quantum communication
– *Vice President, BVP Optica (Optics & Photonics Society), BVCOE*, 2025 – 2026
– *Student Research Lead, Neurogenomo Research Initiative, BVCOE*, 2026 – Present – coordinated interdisciplinary student research projects

References

– **Dr. Neel Kanth Kundu** – Assistant Professor, Indian Institute of Technology Delhi Mail
– **Mr. Rajesh Kumar** – Scientist 'E', Scientific Analysis Group, Defence Research & Development Organisation. Ministry of Defence, Government of India Mail
– **Dr. Yugnanda Malhotra** – Dean R&D, Assistant Professor, Bharati Vidyapeeth's College of Engineering Mail